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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I Semester B. E. Examinations May -2023

Common to AS / CH / IM / ME

CLASSICAL PHYSICS FOR ENGINEERS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer SIX full questions from Part B. In Part B question number 2 and 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.
3. Handbook of Physics is allowed.

PART-A

1	1.1	What are the maximum and minimum values of Poisson's ratio for elastic body?	01
	1.2	What is the value of stress at the central axis of a solid cylinder?	01
	1.3	Where is critical damping observed in automobiles?	01
	1.4	What is forced oscillation?	01
	1.5	State first law of thermodynamics.	01
	1.6	How friction generates irreversibility in a thermodynamic process?	01
	1.7	Give an example for non-Newtonian fluid.	01
	1.8	What is cavitation?	01
	1.9	Name the parameters that can be measured from surface characterization.	01
	1.10	Mention any two disadvantages of AFM.	01

PART-B

2	a	Define spring constant. With neat labeled diagrams, derive an expression for 2 springs in series and in parallel. 2 springs of natural frequency $1Hz$ and $1.2Hz$ are attached in series to a mass of $1kg$. Calculate the effective frequency of oscillations.	10
	b	A circular plate of mass $3.5kg$ and diameter $0.10metre$ is suspended by a wire which passes through its centre. Find the period of angular oscillations for small displacement if the torque required per unit twist of the wire is $3.5 \times 10^{-3} N-m/radian$.	04
3	a	Show that shearing strain is equivalent to sum of compression and elongation strain of half its magnitude and shearing stress is equivalent to tensile stress.	10
	b	Calculate the shearing modulus of a cubic block of size $20cm$ whose top layer displaces by $1.5mm$ under the action of a tangential force of $10Mega-Newton$.	04
		OR	
4	a	What is Young's modulus? Derive an expression to determine the Young's modulus of a given material by single cantilever method and deduce the equation for end-point depression beams for rectangular and circular cross-section.	10
	b	A load of $4kg$ produces an extension of $1mm$ in a wire of $4m$ in length and $1mm$ in diameter. Calculate the Young's modulus of the wire.	04

5	a	Explain Zeroth law of thermodynamics with neat diagram. Discuss the construction and working of Six's maximum and minimum thermometer.	10
	b	When the valve of a sealed bottle is opened, atmospheric air rushes in with a process of $101.325kPa$ and $0.3m^3$ of air is measured at atmospheric pressure. Calculate the work done by air.	04
		OR	
6	a	Define adiabatic expansion of gas. Derive an expression for the adiabatic work done by an external force on the ideal gas by deriving the adiabatic equation of state.	10
	b	A stationary mass of gas is compressed without friction from an initial state of $0.34m^3$ and $0.205MPa$ to a final state of $0.18m^3$ and $0.205MPa$, the pressure remaining constant during the process. There is a transfer of $38.7kJ$ of heat from the gas during the process. How much does the internal energy of the gas change?	04
7	a	What are ultrasonic waves? Explain principle, construction and working of ultrasonic interferometer with diagram.	10
	b	Derive Newton's law of viscosity.	04
		OR	
8	a	Discuss no slip condition in fluids. Write a note on laminar and turbulent flows. Explain how the transition of laminar to turbulent flow depends on flow velocity using Reynolds number.	10
	b	A $0.7mm$ diameter glass tube is inserted into water at 20^0C in a cup. Determine the capillary rise of water in the tube. Given the surface tension of water at 20^0C is $0.073N/m$. Given density of water is $1000kg/m^3$. Assume contact angle for water is 0^0.	04
9	a	Describe the construction, working and analysis of universal testing machine.	08
	b	What is indentation? Write a note on Brinell hardness testing.	06
		OR	
10	a	Explain the principle, construction and working of scanning electron microscope with neat diagram. Mention its applications.	09
	b	Discuss the advantages and disadvantages of transmission electron microscope.	05
11		By constructing a ray diagram, explain the apparatus required and procedure involved to find the wavelength of a laser beam by diffraction technique and calculate the wavelength of a laser beam from the data given below. Given the grating constant is $5.08 \times 10^{-5}m$ and the distance between grating and screen is $100cm$.	20

Diffraction order (n)	*Distance $2x_n$*
1	2.6
2	5.4
3	8.2
4	10.8
5	13.8
6	16.4